

Introduction to Git and GitHub



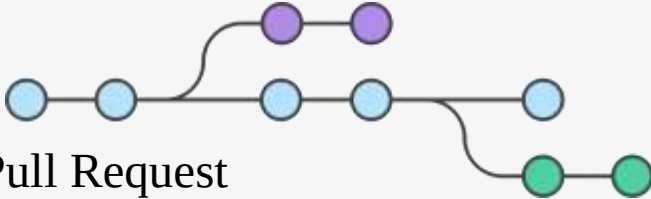
Speakers:
Susan Ghimire
Panas Tiwari

Table of contents

1. Forking



2. Branching



3. Pull Request



Table of contents

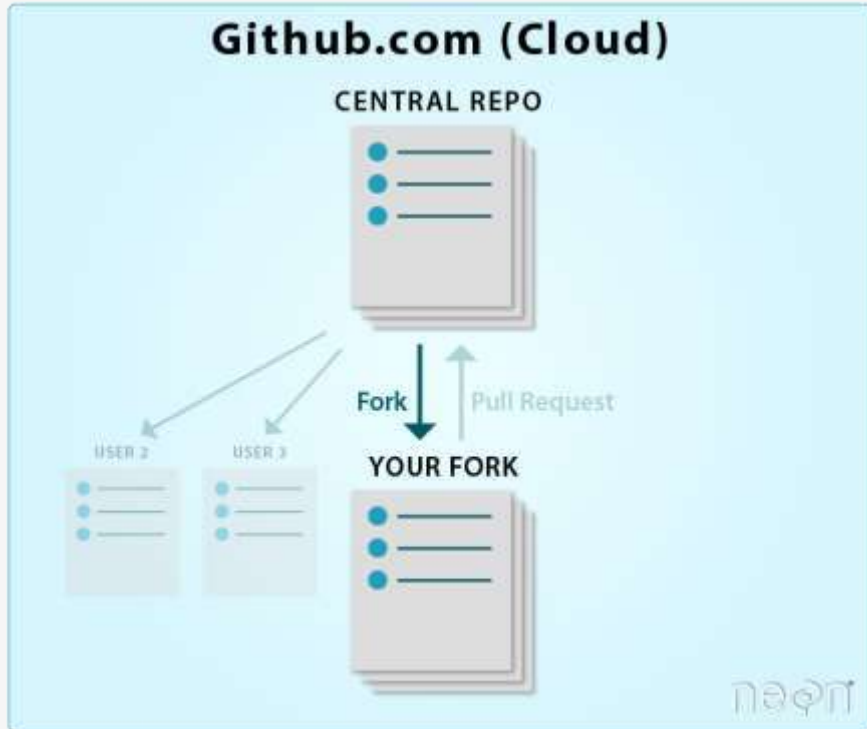
4. Merge Conflicts



5. GitHub issues and ways of assigning them

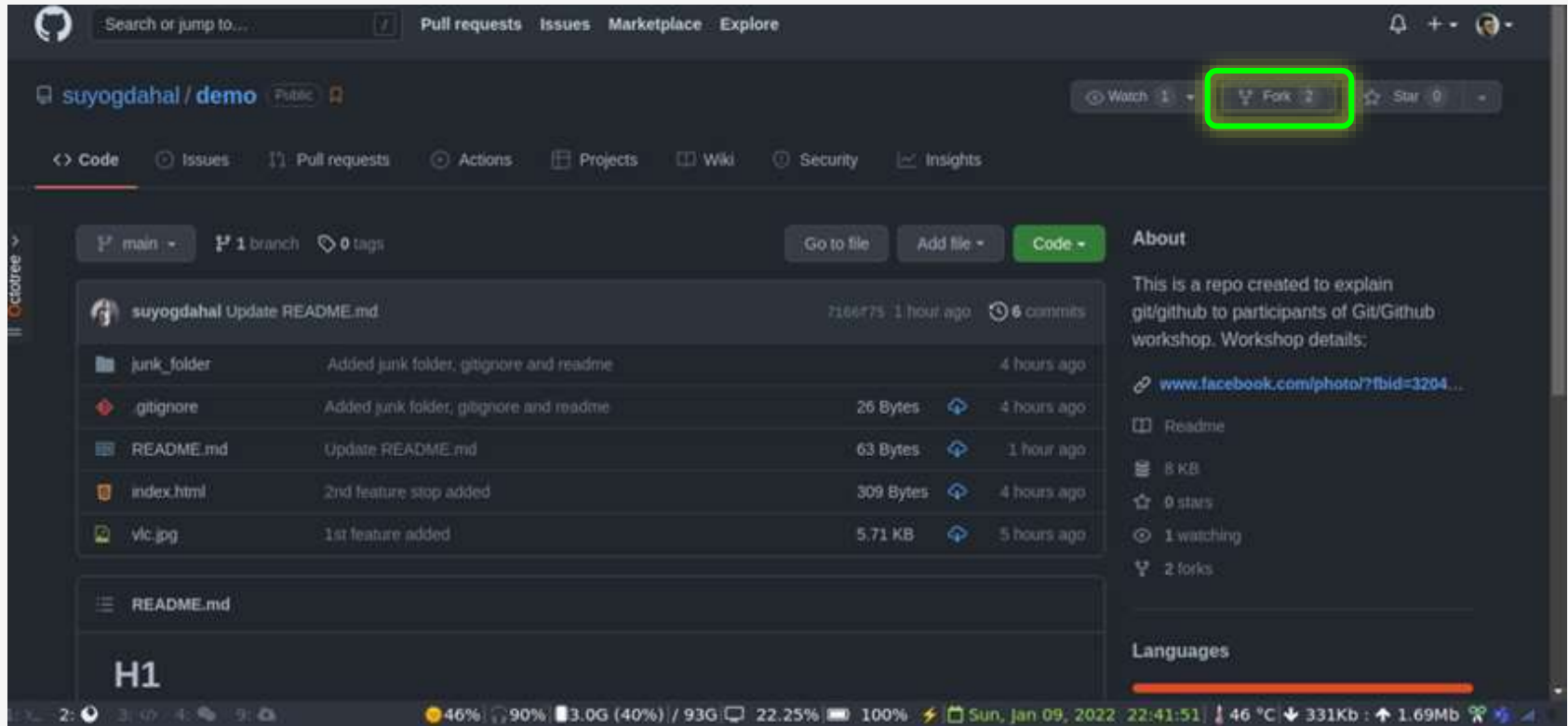


What is Forking?

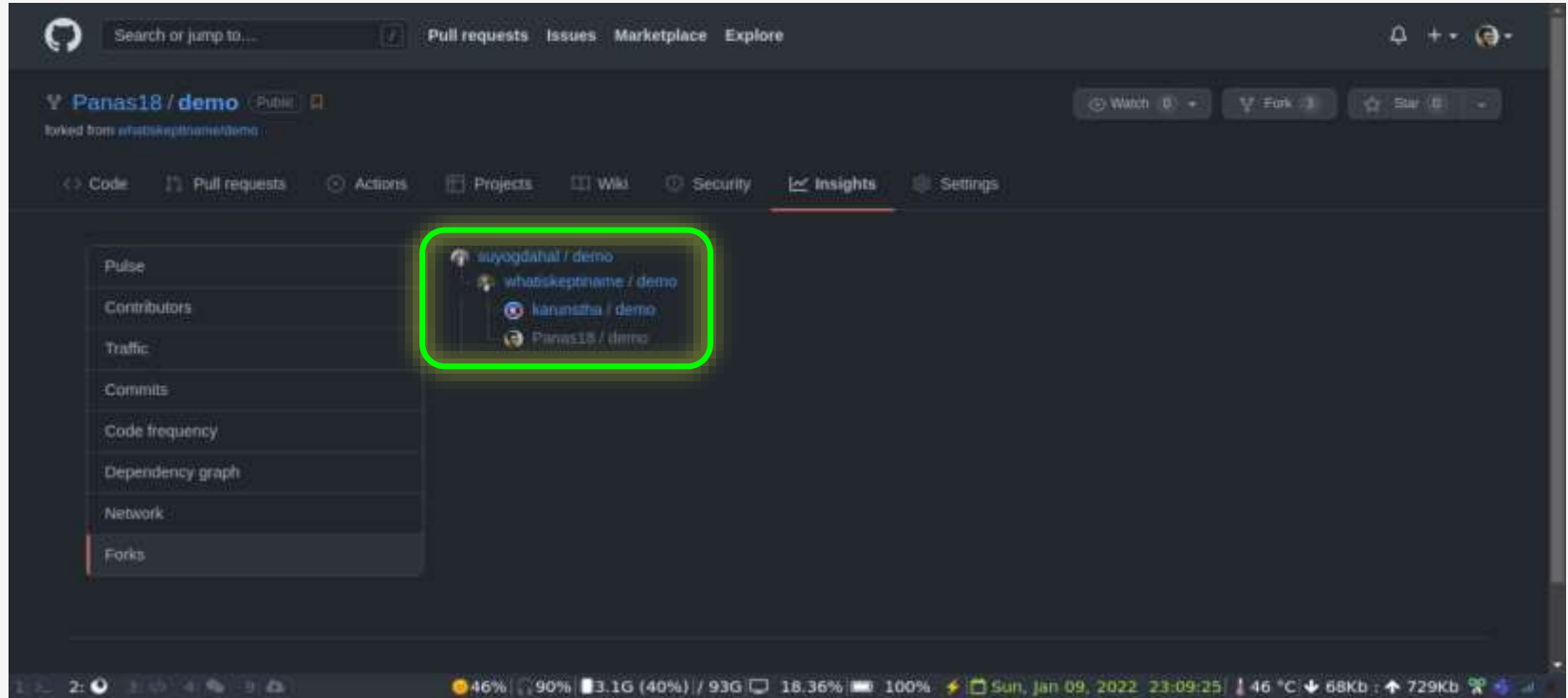


- Forking is a git clone operation executed on a server copy of a projects repo.
- Forking creates your own copy of a repository in a remote location (for example, GitHub).
- Your own copy means that you will be able to contribute changes to your copy of the repository without affecting the original repository.

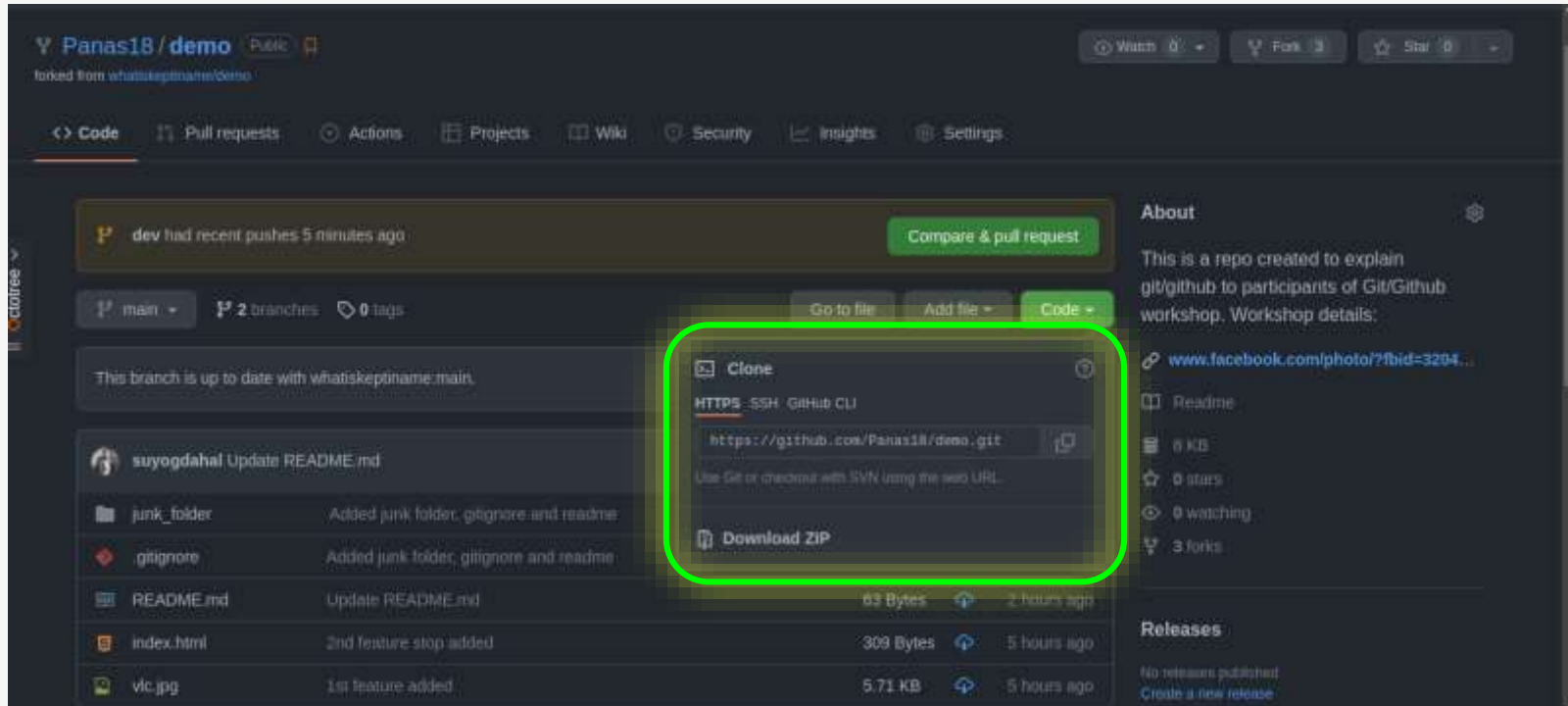
Step 1 : Fork from main repository



Step 2 : Fork from team leader



Step 3 : Git Clone



The screenshot shows the GitHub interface for a repository named 'Panas18/demo'. The repository is public and was forked from 'whatiskptname/demo'. The main branch is 'main', and there are 2 branches and 0 tags. A commit by 'suyogdahal' is shown, titled 'Update README.md', with a list of files: 'junk_folder', '.gitignore', 'README.md', 'index.html', and 'vlc.jpg'. A 'Clone' modal is open, showing the 'HTTPS - SSH - GitHub CLI' options and the URL 'https://github.com/Panas18/demo.git'. The 'Download ZIP' option is also visible. The right sidebar contains an 'About' section with a description of the repository and a 'Releases' section with a link to 'Create a new release'.

Repository: Panas18/demo (Public)

forked from whatiskptname/demo

Navigation: Code, Pull requests, Actions, Projects, Wiki, Security, Insights, Settings

Branches: main (2 branches, 0 tags)

Commit: suyogdahal Update README.md

Files:

File	Commit Message	Size	Time
junk_folder	Added junk folder, gitignore and readme	63 Bytes	2 hours ago
.gitignore	Added junk folder, gitignore and readme	309 Bytes	5 hours ago
README.md	Update README.md	5.71 KB	5 hours ago
index.html	2nd feature stop added		
vlc.jpg	1st feature added		

Clone Options:

- Clone (HTTPS - SSH - GitHub CLI)
- URL: <https://github.com/Panas18/demo.git>
- Download ZIP

About: This is a repo created to explain git/github to participants of Git/Github workshop. Workshop details: www.facebook.com/photo/?fbid=3294...

Releases: No releases published. [Create a new release](#)

\$ git clone <link>

- This command is used to point to an existing repo and make a clone or copy of that repo at in a new directory, at another location.

```
(base) [git_workshop/git] git clone git@github.com:Panas18/demo.git
Cloning into 'demo'...
Enter passphrase for key '/home/panas/.ssh/id_rsa':
remote: Enumerating objects: 20, done.
remote: Counting objects: 100% (20/20), done.
remote: Compressing objects: 100% (15/15), done.
remote: Total 20 (delta 4), reused 14 (delta 2), pack-reused 0
Receiving objects: 100% (20/20), 8.10 KiB | 4.05 MiB/s, done.
Resolving deltas: 100% (4/4), done.
(base) [git_workshop/git]
```

What is Git branching?



- Branching is the practice of creating copies of programs or objects in development to work in parallel versions, retaining the original and working on the branch or making different changes to each.
- A branch is essentially is a unique set of code changes with a unique name
- Edit code on new branch without impacting the main branch.
- Common examples: Stable, Testing etc

Step 4 : Create New Branch

```
(base) [git_workshop/demo] git branch dev
```

\$ git branch

- This command is used to to list all the branches

```
dev  
* main  
(END)
```

Step 5 : Change Branch

```
(base) [git_workshop/demo] git checkout dev  
Switched to branch 'dev'  
(base) [git_workshop/demo]
```

Step 6 : Add features inside new branch

```
16 <html>
15 <head>
14 <title> GIT week1 </title>
13 </head>
12
11 <body>
10 <center>
9
8
7 <h3> VLC Media Player </h3>
6
5 
4 <br><br>
3
2 <h2> i) 1st feature : {{PLAY}} </h2>
1 <h2> ii) 2nd feature : {{STOP}} </h2>
17 <h2> iii) 3rd feature : {{FORWARD}} </h2>
1
2 </center>
3 </body>
4
5 </html>
```

index.html 17,5-11 All

-- VISUAL LINE --

1: 46% 90% 3.2G (42%) / 93G 22.34% 100% Sun, Jan 09, 2022 23:43:18 47 °C 68Kb 550Kb

Step 7 : Commit changes

```
(base) [git_workshop/demo] git add index.html
(base) [git_workshop/demo] git commit -m "3rd feature added"
[dev 3194ebb] 3rd feature added
1 file changed, 1 insertion(+)
(base) [git_workshop/demo]
(base) [git_workshop/demo]
(base) [git_workshop/demo]
(base) [git_workshop/demo]
```

Step 8 : Git merge

```
(base) [git_workshop/demo] git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
(base) [git_workshop/demo] git merge dev
Updating 7166f75..3194ebb
Fast-forward
 index.html | 1 +
 1 file changed, 1 insertion(+)
(base) [git_workshop/demo] git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
(base) [git_workshop/demo] git push
Enter passphrase for key '/home/panas/.ssh/id_rsa':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 329 bytes | 329.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:Panas18/demo.git
 7166f75..3194ebb  main -> main
```

Any Questions??

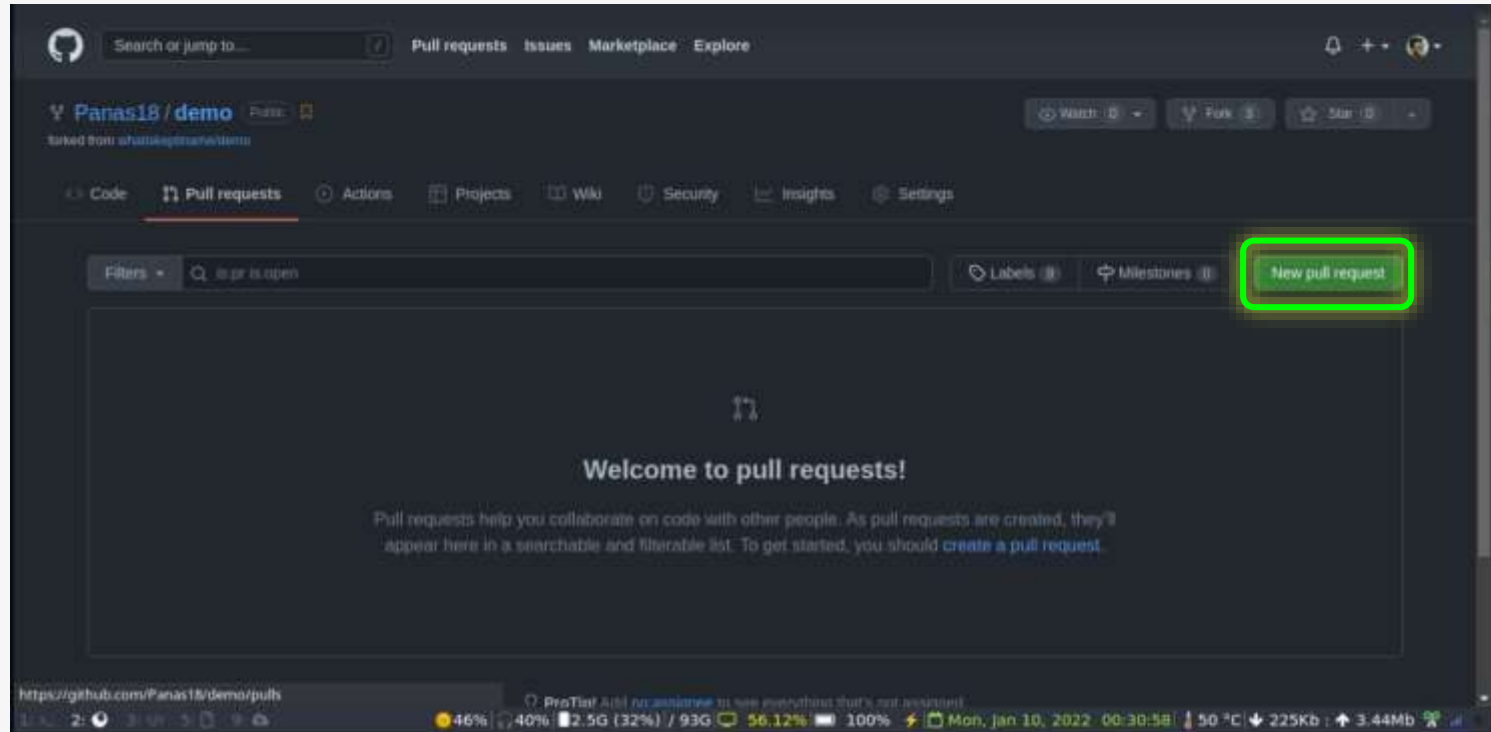


What is Pull Request?

- A pull request is a method of submitting contributions to an open development project.
- A pull request occurs when a developer asks for changes committed to an external repository to be considered for inclusion in a project's main repository.



Creating pull request



whatiskeptname / demo

forked from suyogdhatname

Code Pull requests Actions Projects Wiki Settings

main 2 branches 0 tags

This branch is 7 commits ahead of suyogdhat:main. Fetch upstream

whatiskeptname Merge pull request #1 from Panas181man 5c1d812 7 minutes ago 13 commits

junk_folder	Added junk folder, gitignore and readme	0 hours ago
.gitignore	Added junk folder, gitignore and readme	26 Bytes 6 hours ago
README.md	Update README.md	63 Bytes 3 hours ago
bl.py	main dump	55 Bytes 2 hours ago
index.html	Merge branch 'main' into main	464 Bytes 7 minutes ago
Vic.jpg	1st feature added	5.71 KB 7 hours ago

README.md

H1

H2

This is a repo created to explain git/github to participants of Git/Github workshop. Workshop details:

www.facebook.com/photo/?fbid=3204...

0 stars
0 watching
3 forks

HTML 88.4% Python 10.6%

whatiskeptiname / demo
forked from suyogdahalidemo

Watch Fork Star 0

Code Pull requests 1 Actions Projects Wiki 0 Settings

Filters is:pr is:open sort:updated-desc Labels 0 Milestones 0 New pull request

Clear current search query, filters, and sorts

1 Open 1 Closed Merged Author Label Projects Reviews Assignee Sort

added new feature
#2 opened 3 minutes ago by karunsth

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Merge Conflict

A merge conflict is an event that occurs when Git is unable to automatically resolve differences in code between two commits.

In real world merge conflict happens because of following reasons:

- Same line of code is changed in two branches.
- A given file is changed in one branch and deleted in another branch
- Same file is added twice in two different branch but the content of the file is different

added new feature #2

Edit <> Code

Open what I kept in main ← karuntha:main

Conversation 0 Commits 1 Checks 1 Files changed 1

+1 -0



karuntha 3 minutes ago

First-time contributor

No description provided.

add new feature

asD7ada

Add more commits by pushing to the **main** branch on **karunthaidemo**.



This branch has conflicts that must be resolved

Use the web editor or the command line to resolve conflicts.

Conflicting files

index.html

Resolve conflicts

Merge pull request

Convert to draft or view command line instructions



Write

Preview



H B I E <> Link List Table Image Video Poll Giphy GIF

Leave a comment

Close pull request

Comment

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Reviewers - review now

Sit in progress? Convert to draft

Assignees - assign yourself

Labels

Projects

Milestone

Notifications

Customize

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1 participant



Lock conversation

added new feature #2

Resolving conflicts between [karunitha:main](#) and [whatiskeyinake:main](#) and committing changes → [karunitha:main](#)

1 conflicting file

index.html
index.html

index.html

Accept Current Change Accept Incoming Change Accept Both Changes

```
1 <!--- Karunitha's Change -->
2 <html>
3 <head>
4 <title> GIT week1 </title>
5 </head>
6
7 <body>
8 <center>
9
10
11 <h3> VLC Media Player </h3>
12
13 
14 <br><br>
15
16 <h2> 1) 1st Feature : {{PLAY}} </h2>
17 <h2> 11) 2nd Feature : {{STOP}} </h2>
18 <h2> 11) 3rd Feature : {{NEXT}} </h2>
19
20 </center>
21 </body>
22
23 </html>
24 <!--- Karunitha's Change -->
25
26 <!--- whatiskeyinake's Change -->
27 <html>
28 <head>
29 <title> GIT week1 </title>
30 </head>
31
32 <body>
33 <center>
34
35
36 <h3> VLC Media Player </h3>
37
38 
39 <br><br>
40
41 <h2> 1) 1st Feature : {{PLAY}} </h2>
42 <h2> 11) 2nd Feature : {{STOP}} </h2>
43 <h2> 111) 3rd Feature : {{RANDOM}} </h2>
```

added new feature #2

Resolving conflicts between [karimstra:main](#) and [whatiskpt:show:main](#) and committing changes → [karimstra:main](#)

1 conflicting file

index.html
index.html

index.html

1 conflict

Prev

Next

⚙

Mark as resolved

```
1  <html>
2  <head>
3    <title> GIT week1 </title>
4  </head>
5
6  <body>
7    <center>
8
9
10     <h3> VLC Media Player </h3>
11
12     
13     <br><br>
14
15     <h2> 1) 1st Feature : {{PLAY}} </h2>
16     <h2> 11) 2nd Feature : {{STOP}} </h2>
17     <h2> 11) 3rd Feature : {{NEXT}} </h2>
18
19   </center>
20 </body>
21
22 </html>
23 </html>
24
25 <head>
26   <title> GIT week1 </title>
27 </head>
28
29 <body>
30   <center>
31
32
33   <h3> VLC Media Player </h3>
34
35   
36   <br><br>
37
38   <h2> 1) 1st Feature : {{PLAY}} </h2>
39   <h2> 11) 2nd Feature : {{STOP}} </h2>
40   <h2> 111) 3rd Feature : {{FORWARD}} </h2>
41   
```

added new feature #2

Resolving conflicts between karunatha:main and whatiskept:main and committing changes -> karunatha:main

Commit merge

1 conflicting file

index.html

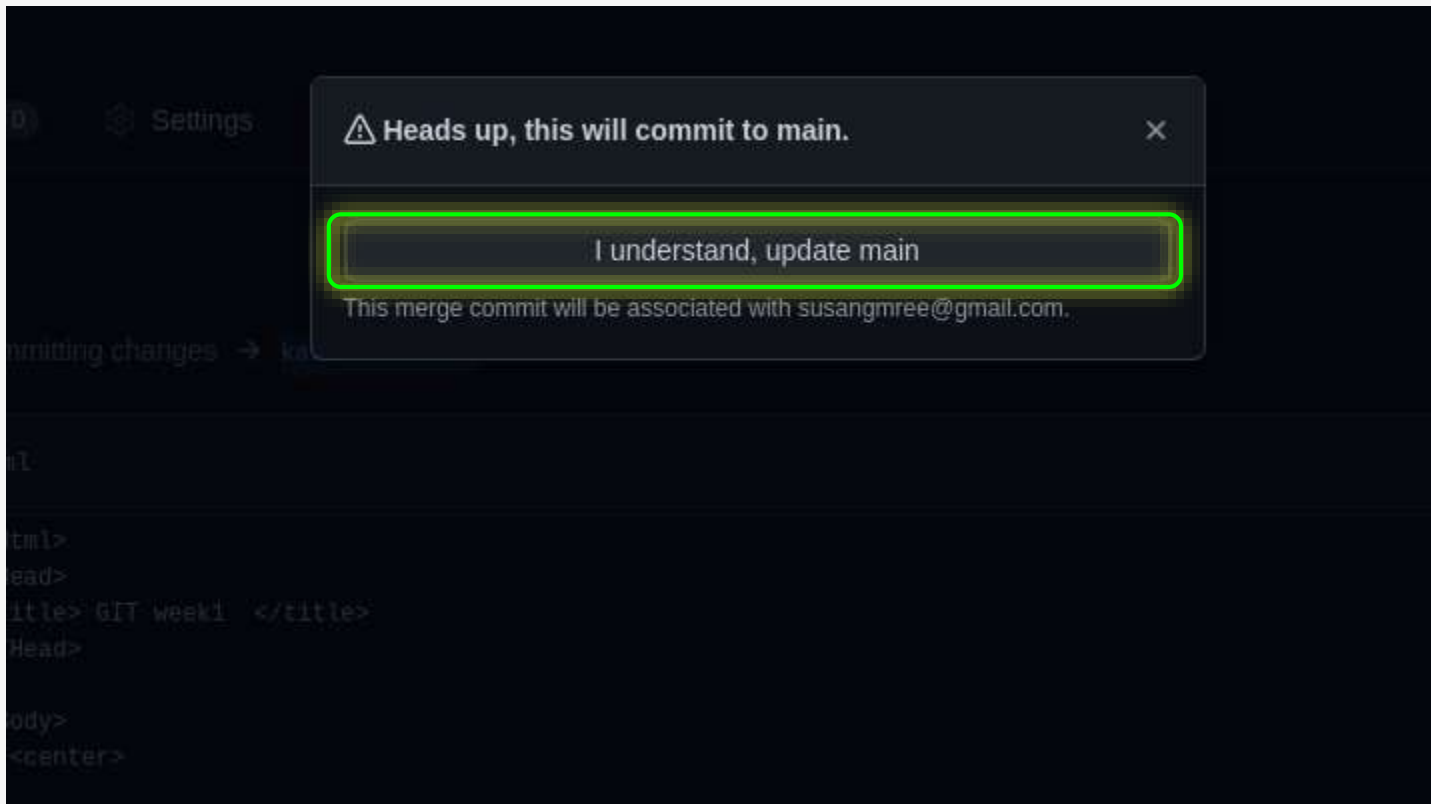
index.html



index.html

```
1 <html>
2 <head>
3 <title> GIT week1 </title>
4 </head>
5
6 <body>
7 <center>
8
9
10 <h3> VLC Media Player </h3>
11
12 
13 <br><br>
14
15 <h2> 1) 1st feature : {{PLAY}} </h2>
16 <h2> 1) 2nd feature : {{STOP}} </h2>
17 <h2> 1) 3rd feature : {{NEXT}} </h2>
18
19 </center>
20 </body>
21
22 </html>
23 <html>
24
25 <head>
26 <title> GIT week1 </title>
27 </head>
28
29 <body>
30 <center>
31
32
33 <h3> VLC Media Player </h3>
34
35 
```

✓ Resolved



added new feature #2

[Edit](#)[Code](#)[Open](#)[whatiskptname/main](#) ← [karuntha/main](#)[Conversation](#) (0)[Commits](#) (2)[Checks](#) (0)[Files changed](#) (1)[+23](#) [-1](#) [0](#) [0](#) [0](#) [0](#)**karuntha** 25 minutes ago[First time contributor](#)

No description provided.

[Reviewers](#) – review now[Suggestions](#)[whatiskptname](#)[Request](#)

Still in progress? Convert to draft

[Assignees](#) – assign yourself[Labels](#)[Projects](#)[Milestone](#)[Notifications](#)[Customize](#)

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[Subscribe](#)[2 participants](#)[Lock conversation](#)

karuntha and others added 2 commits 26 minutes ago



added new feature

a5d79da



Merge branch "main" into main

[Verified](#)

18127cd

Add more commits by pushing to the **main** branch on **karuntha/demo**.

**This branch has no conflicts with the base branch**[Merge pull request](#)[Convert to draft](#) or [view command line instructions](#).[Write](#)[Preview](#)

Leave a comment

[Close pull request](#)[Comment](#)

Remember, contributions to this repository should follow our [GitHub Community Guidelines](#).

added new feature #2

[Edit](#)[Code](#)[Open](#)[whatiskptname/main](#) ← [karuntha/main](#)[Conversation](#) 0[Commits](#) 2[Checks](#) 0[Files changed](#) 1

+23 -1

**karuntha** 26 minutes ago

First-time contributor

No description provided.**karuntha** and others added 2 commits 27 minutes ago

added new Feature

a5d7dda



Merge branch 'main' into main

Verified

1612f08

Add more commits by pushing to the **main** branch on **karuntha/demo**.

Merge pull request #2 from karuntha/main

Added a new feature.



susangnree@gmail.com

Choose which email address to associate with this commit

Confirm merge

Cancel

Reviewers – review now



Suggestions

**whatiskptname**[Request](#)

Still in progress? Convert to draft

Assignees – assign yourself



Labels



Projects



Milestone



Notifications

[Customize](#)

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2 participants

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added new feature #2

Merged

by whatiskeptiname • whatiskeptiname:feat ← karuntha:main now

Conversation 0

Commits 2

Checks 0

Files changed 1

+23 -1



karuntha 26 minutes ago

No description provided.



karuntha and others added 2 commits 27 minutes ago



added new feature

a5d7dd



Merge branch 'feat' into main

Verified

1812fcs



whatiskeptiname merged commit 342a5a6 into whatiskeptiname:main now

Revert



Write

Preview



≡ <> 🔗 ≡ ≡ ≡ ≡ ≡ ≡ ≡ ≡

Leave a comment

Comment

Remember, contributions to this repository should follow our [GitHub Community Guidelines](#).

ProTip! Add comments to specific lines under **Files changed**.

Reviewers – review now

Assignees – assign yourself

Labels

Projects

Milestone

Notifications

Customize

You're receiving notifications because you modified the open/close state.

Unsubscribe

2 participants

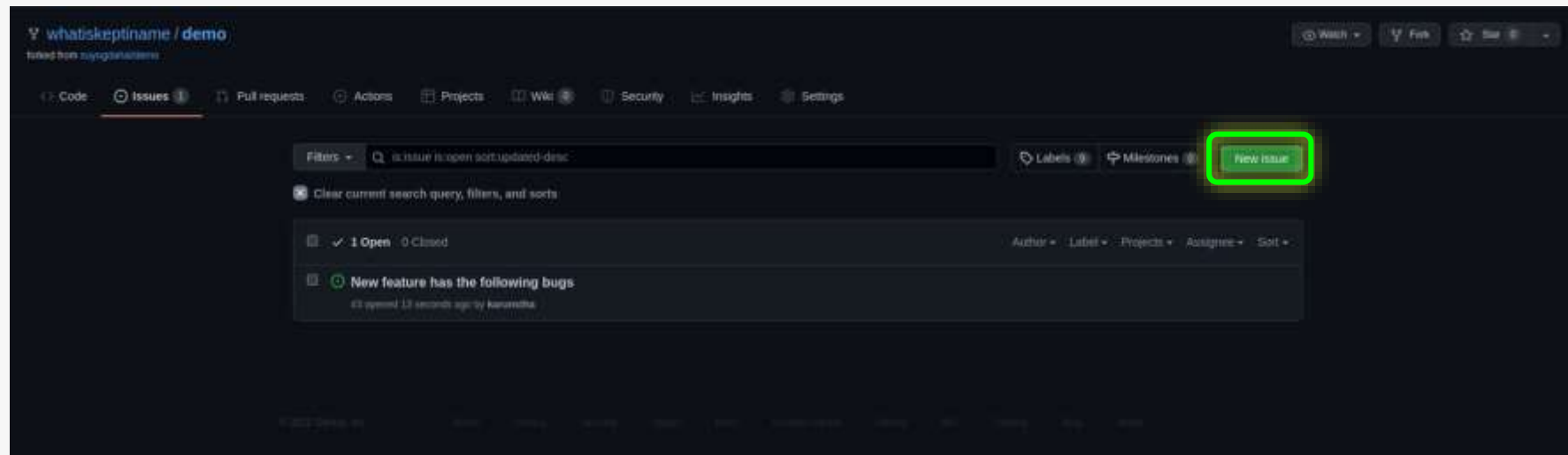


Lock conversation



GitHub Issues and ways of assigning them

- GitHub Issues is a tracking tool that is integrated with your GitHub repository.
- Issues let you track your work on GitHub, where development happens.
- Github issues is used to:
 - Track ideas
 - Feedback
 - Assign task
 - Report bugs



Code Issues Pull requests Actions Projects Wiki Security Insights Settings

New feature has the following bugs #3

Open 1 comment

karunitha 2 minutes ago

1. Bug 1

whatiskeptiname added the **invalid** label now

whatiskeptiname now

This is not a bug but a feature.

Write Preview

Labels & comments

Close Issue

Comment

Assignees - assign yourself

Labels

invalid

Projects

Milestone

Linked pull requests

Notifications

You're receiving notifications because you commented.

Unsubscribe

2 participants

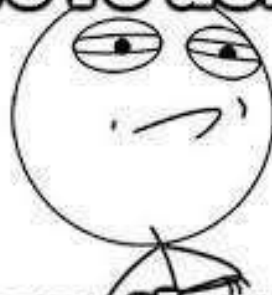
Lock conversation

Pin issue

Transfer issue

Delete issue

We're done.



Questions?

Assignment

If you're facing any difficulties:

Contact us at:

Facebook: [Kec Robotics Club](#) [Kec I.T. Club](#)

Mail: rbtkec@gmail.com

kecktm.it.club@gmail.com



